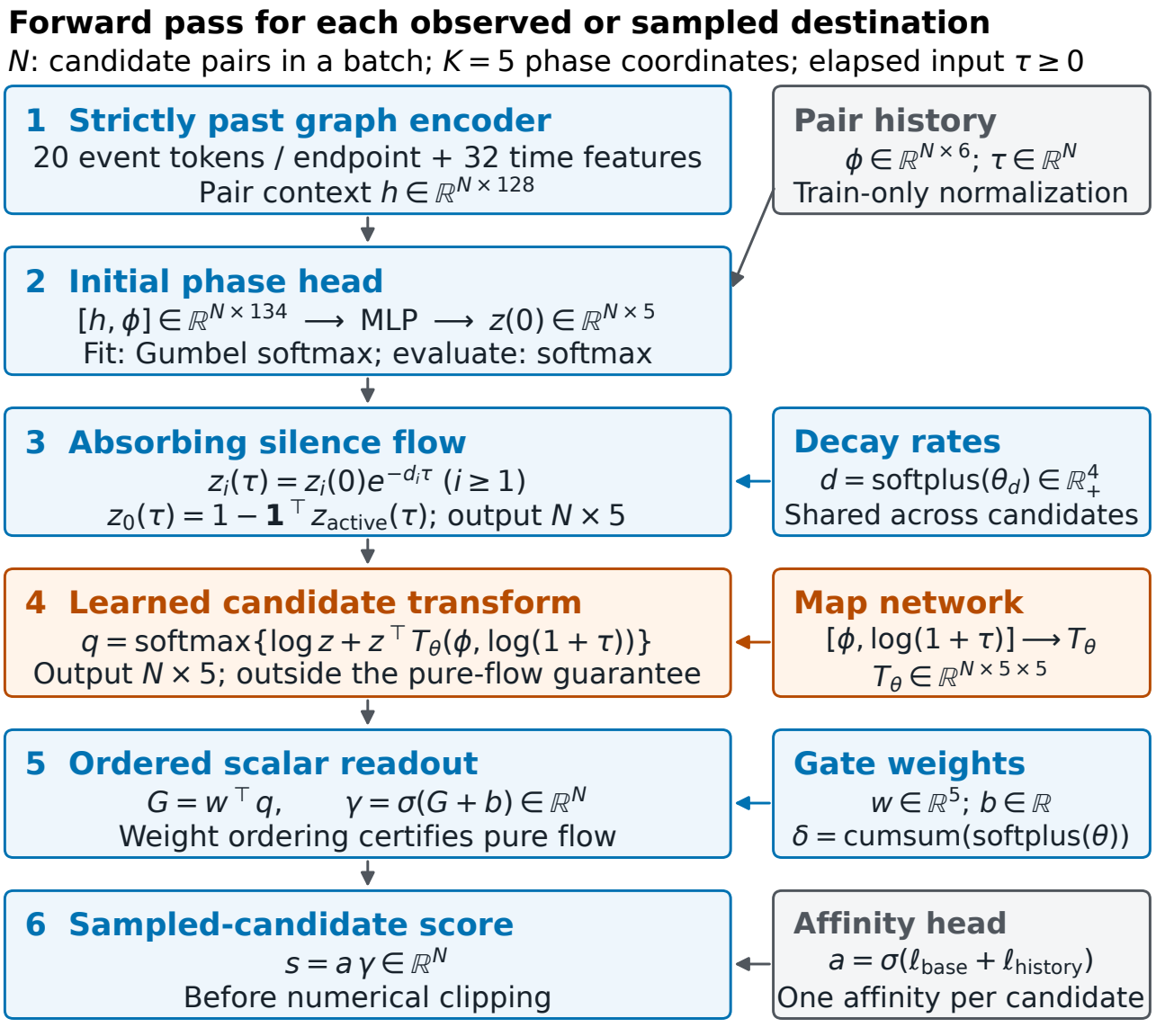
Forward pass for each observed or sampled destination
N: candidate pairs in a batch; K = 5 phase coordinates; elapsed input τ ≥ 0
1 Strictly past graph encoder
20 event tokens / endpoint + 32 time features
Pair context h ∈ ℝ^{N×128}
Pair history
φ ∈ ℝ^{N×6}; τ ∈ ℝ^N
Train-only normalization
2 Initial phase head
[h, φ] ∈ ℝ^{N×134} ⟶ MLP ⟶ z(0) ∈ ℝ^{N×5}
Fit: Gumbel softmax; evaluate: softmax
3 Absorbing silence flow
z_i(τ) = z_i(0)e^{−d_i τ} (i ≥ 1)
z_0(τ) = 1 − 1^⊤ z_active(τ); output N × 5
Decay rates
d = softplus(θ_d) ∈ ℝ^4_+
Shared across candidates
4 Learned candidate transform
q = softmax{log z + z^⊤ T_θ(φ, log(1 + τ))}
Output N × 5; outside the pure-flow guarantee
Map network
[φ, log(1 + τ)] ⟶ T_θ
T_θ ∈ ℝ^{N×5×5}
5 Ordered scalar readout
G = w^⊤ q, γ = σ(G + b) ∈ ℝ^N
Weight ordering certifies pure flow
Gate weights
w ∈ ℝ^5; b ∈ ℝ
δ = cumsum(softplus(θ))
6 Sampled-candidate score
s = a γ ∈ ℝ^N
Before numerical clipping
Affinity head
a = σ(ℓ_base + ℓ_history)
One affinity per candidate
Silence check: freeze context, state and affinity; vary elapsed input.